[illegible]

III. Abstract Integration. 1 Abstract measures

Def III. 1.1. $\mathfrak{M} \subseteq \mathcal{P}(X)$ is a σ-Algebra in X if

(i) $X \in \mathfrak{M}$

(ii) If $A \in \mathfrak{M}$, then $A^c \in \mathfrak{M}$

(iii) If $A = \bigcup_{n=1}^{\infty} A_n$ $A_n \in \mathfrak{M}$ then $A \in \mathfrak{M}$

Then X is called a measurable ~~set~~ space. Elements of $\mathfrak{M}$ ~~is~~ are measurable sets

Def III. 1. 2. $f: X \to Y$ X measurable space. Y topological space

f is measurable iff $f^{-1}(V)$ is measurable all open sets V

Let X be a measurable space

ef III.1.3. A measure $\mu: \mathfrak{M} \longrightarrow [0,+\infty]$. with the following property:

$E_i \in \mathfrak{M}$. Then $\mu(\bigcup_{n=1}^{\infty} E_n) = \sum_{n=1}^{\infty} \mu(E_n)$

ef III.1.4. Let X be a set, An exterior measure $\mu_*: \mathcal{P}(X) \longrightarrow [0,+\infty]$.

that satisfies: (i) $\mu_*(\phi) = 0$

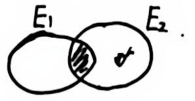
(ii) If $E_1 \subseteq E_2$ then $\mu_*(E_1) \le \mu_*(E_2)$

(iii) ~~If~~ $\forall \{E_k\}_{k\ge1}$. ~~then~~ we have $\mu_*(\cancel{\sum_{k\ge1}} \bigcup_{k\ge1} E_k) \le \sum_{k\ge1} \mu_*(E_k)$

ef III.1.5. A set E in X is measurable iff $\mu_*(A) = \mu_*(E \cap A) + \mu_*(E^c \cap A)$. $\forall A \subseteq X$.

hm III.1.6 (Carathéodory theorem) Given an exterior measure μ_* on a set X, the ~~restriction~~ collection $\mathfrak{M}$ of measurable sets forms a σ-algebra. Moreover, $\mu_*|_{\mathfrak{M}}$ is a measure

roof: Clearly 1) ϕ, $X \in \mathfrak{M}$



2) If $E \in \mathfrak{M}$. then $E^c \in \mathfrak{M}$

3) If $E_i \in \mathfrak{M}$, then $\mu_*(A) = \mu_*(E_2 \cap A) + \mu_*(E_2^c \cap A)$

$$= \mu_*(E_1 \cap E_2 \cap A) + \mu_*(E_1^c \cap E_2 \cap A) + \mu_*(E_1^c \cap E_2^c \cap A) + \mu_*(E_1 \cap E_2^c \cap A$$

$E_1 \cup E_2 = (E_1 \cap E_2) \cup (E_1 \cap E_2^c) \cup (E_1^c \cap E_2)$

$$\equiv \mu_*((E_1 \cup E_2) \cap A) + \mu_*((E_1 \cup E_2)^c \cap A).$$

$\Rightarrow E_1 \cup E_2 \in \mathfrak{M}$. Let $G_n = \bigcup_{i=1}^{n} E_i$ $E = \bigcup_{n\ge1} E_n$

By induction: ~~[illegible] $\subseteq \mu_*(E_n$~~

$$\mu_*(G_n \cap A) = \mu_*(E_n \cap (G_n \cap A)) + \mu_*(E_n^c \cap (G_n \cap A))$$

$$= \mu_*(E_n \cap A) + \mu_*(G_{n-1} \cap A) = \sum_{j=1}^{n} \mu_*(E_j \cap A)$$

$$\Rightarrow \mu_*(A) = \mu_*(G_n \cap A) + \mu_*(G_n^c \cap A) = \sum_{j=1}^{n} \mu_*(E_j \cap A) + \mu_*(G_n^c \cap A)$$

$$\ge \sum_{j=1}^{n} \mu_*(E_j \cap A) + \mu_*(G^c \cap A) \quad n \to \infty \ge \sum_{j=1}^{\infty} \mu_*(E_j \cap$$

$+\mu_*(G^c \cap$

$$\ge \mu_*(G \cap A) + \mu_*(G^c \cap A) \square$$

T. (Exe of

2. Metric exterior measures (Now we assume that X is a metric space)

Def II.2.1. X is a metric space. The Borel σ-algebra $\mathcal{B}_X$ is the smallest σ-algebra of sets in X that contains the open sets in X. Elements in $\mathcal{B}_X$ are called Borel sets

Def II.2.2. An exterior measure μ_* on X is a metric exterior measure if it satisfies $\mu_*(A \cup B) = \mu_*(A) + \mu_*(B)$ whenever $d(A,B) > 0$

Thm II.2.3 μ_* is a metric exterior measure on X, then the Borel sets in X are measurable. Hence $\mu_*|_{\mathcal{B}_X}$ is a measure.

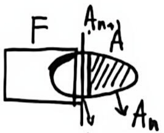
Proof: We just need to show that closed sets are measurable

Let F be closed. $A \subseteq X$. $\mu_*(A) < \infty$. Let $A_n = \{x \in F^c \cap A \mid d(x,F) > \frac{1}{n}\}$.

Then $A_n \subseteq A_{n+1}$, $F^c \cap A = \bigcup\limits_{n \geq 1} A_n$

And we have $d(F \cap A, A_n) \geq \frac{1}{n}$ thus. $\mu_*(A) \geq \mu_*((F \cap A) \cup A_n)$
$= \mu_*(F \cap A) + \mu_*(A_n)$.

Next we'll show that $\lim\limits_{n \to \infty} \mu_*(A_n) = \mu_*(F^c \cap A)$.



Let $B_n = A_{n+1} \cap A_n^c$., $d(B_{n+1}, A_n) \geq \frac{1}{n(n+1)}$ (If $\exists\, x \in B_{n+1}$, $y \in A_n$ s.t $d(x,y) < \frac{1}{n(n+1)}$
Then $d(y,F) < d(x,y) + d(x,F) < \frac{1}{n(n+1)} + \frac{1}{n+1} = \frac{1}{n}$
Contradiction !)

Then $\mu_*(A_{2k+1}) \geq \mu_*(B_{2k} \cup A_{2k-1}) = \mu_*(B_{2k}) + \mu_*(A_{2k-1})$

$\Rightarrow \mu_*(A_{2k+1}) \geq \sum\limits_{j=1}^{k} \mu_*(B_{2j})$ Similarly, $\mu_*(A_{2k}) \geq \sum\limits_{j=1}^{k} \mu_*(B_{2j-1})$. $\Rightarrow \sum \mu_*(B_j)$ converges

Note that $\mu_*(A_n) \leq \mu_*(F^c \cap A) \leq \mu_*(A_n) + \sum\limits_{j=n+1}^{\infty} \mu_*(B_j)$

let $n \to \infty$ $\Rightarrow$ $\mu_*(F^c \cap A) = \lim\limits_{n \to \infty} \mu_*(A_n)$

Thus $\mu_*(A) \geq \mu_*(F \cap A) + \mu_*(F^c \cap A)$ $\square$

Given a metric space X, μ defined on the Borel sets of X is called the Borel measure.

Prop III.2.4. Suppose the Borel measure μ is finite on all balls in X of finite radius. Then for any Borel set E, $\forall \varepsilon > 0$. $\exists\ \mathcal{O}$ open. F closed. s.t. $F \subseteq E \subseteq \mathcal{O}$ $\mu(\mathcal{O}-E)<\varepsilon$, $\mu(E-F)<\varepsilon$

Proof: